LAHORE UNIVERSITY OF MANAGEMENT SCIENCES
Department of Electrical Engineering

AI 501 Mathematics for Artificial Intelligence
Quiz 4 – Solutions

Name: _____

Campus ID: _____

Total Marks: 10
Time Duration: 25 minutes

Question 1 (3 marks)

Consider the bivariate function $f(x, y) = 2x^2 + 3xy + y^2$.

(a) Find the gradient of the function, $\nabla f(x, y)$. [1 mark]

Solution: The gradient is the vector of partial derivatives:

$$\nabla f(x, y) = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} = \begin{pmatrix} 4x + 3y \\ 3x + 2y \end{pmatrix}$$

(b) Find the Hessian matrix of the function, $H_f(x, y)$. [2 marks]

Solution: The Hessian is the matrix of second-order partial derivatives:

$$H_f(x, y) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix} = \begin{pmatrix} 4 & 3 \\ 3 & 2 \end{pmatrix}$$

Question 2 (5 marks)

This question assesses the batch gradient descent update rule for linear regression.

(a) [1 mark] Write down the update rule for a single parameter θ_j in a linear regression model.

Solution: The update rule is:

$$\theta_j \leftarrow \theta_j - \alpha \frac{1}{n} \sum_{i=1}^{n} (\theta_0 + \boldsymbol{\theta}^T \mathbf{x}_i - y_i) x_{ij}$$

(b) [4 marks] Consider a single data point $(\mathbf{x}_1, y_1) = ((2), 5)$ for a model $h(\mathbf{x}) = \theta_0 + \theta_1 x_1$. Given initial parameters $\theta_0 = 1, \theta_1 = 1$ and a learning rate $\alpha = 0.1$, compute the updated values for θ_0 and θ_1 after one step of gradient descent.

Solution: Here, $n = 1$. First, calculate the prediction error:

$$\text{error} = (\theta_0 + \theta_1 x_1 - y_1) = (1 + 1(2) - 5) = 3 - 5 = -2$$

Next, compute the gradients (for $n = 1$, the sum is just the single term):

$$\frac{\partial L}{\partial \theta_0} = \frac{1}{1}(\text{error}) \cdot 1 = -2$$

$$\frac{\partial L}{\partial \theta_1} = \frac{1}{1}(\text{error}) \cdot x_1 = (-2)(2) = -4$$

Finally, apply the simultaneous update rules:

$$\theta_0 \leftarrow \theta_0 - \alpha \frac{\partial L}{\partial \theta_0} = 1 - (0.1)(-2) = 1 + 0.2 = 1.2$$

$$\theta_1 \leftarrow \theta_1 - \alpha \frac{\partial L}{\partial \theta_1} = 1 - (0.1)(-4) = 1 + 0.4 = 1.4$$

The updated parameters are $\theta_0 = 1.2$ and $\theta_1 = 1.4$.

Question 3 (2 marks)

Using Jensen's inequality, $f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta)f(y)$, prove that the linear function $f(x) = a^T x + b$ is convex.

Solution: We need to check if the inequality holds for $f(x) = a^T x + b$.

$$\begin{aligned} f(\theta x + (1-\theta)y) &= a^T(\theta x + (1-\theta)y) + b \\ &= a^T(\theta x) + a^T((1-\theta)y) + b \\ &= \theta(a^T x) + (1-\theta)(a^T y) + (\theta + 1 - \theta)b \\ &= \theta(a^T x) + (1-\theta)(a^T y) + \theta b + (1-\theta)b \\ &= \theta(a^T x + b) + (1-\theta)(a^T y + b) \\ &= \theta f(x) + (1-\theta)f(y) \end{aligned}$$

Since $f(\theta x + (1-\theta)y) = \theta f(x) + (1-\theta)f(y)$, the inequality $\leq$ is satisfied. Therefore, any linear function is convex. (It is also concave).